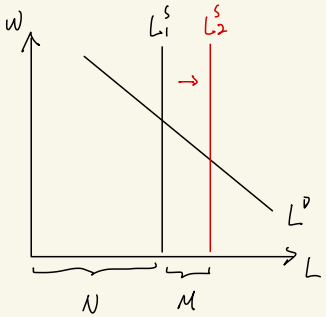
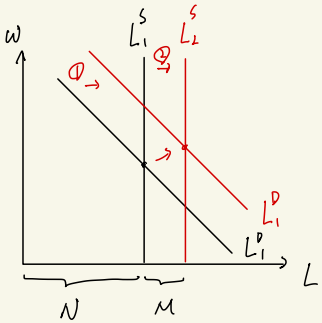
The effect of immigration on wage

What we want to study is the causal effect of immigration on wage



What we see in the data may be ...



Economic growth

↓

Labor demand ↑

↓

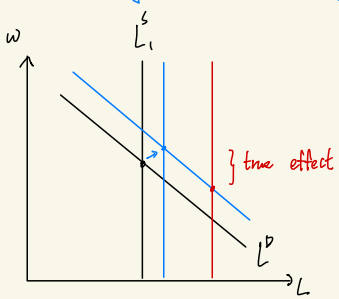
Attract immigrants, L^S ↑ (Selection problem)

How do researchers solve this problem ?

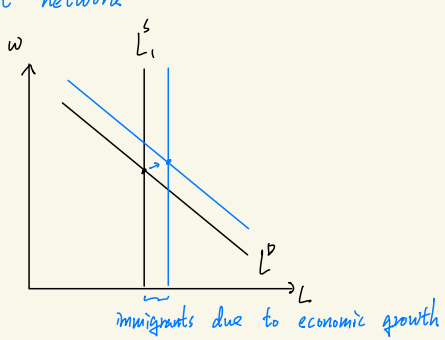
→ Find a variable that could affect immigration but "unrelated" to labor demand

We call this variable an instrumental variable.

e.g. historical immigrant network

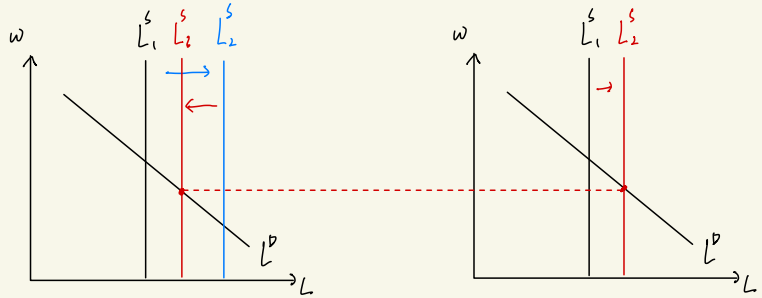


A : with network



B : without network

However, immigrants may move from A to B because the wage in B becomes higher.



A B

If so, we would see immigration has no effect on wage.

Besides, native workers and immigrants may not be perfect substitutes.

- Elasticity of substitution

$$\ln\left(\frac{L_N}{L_F}\right) = \text{constant} - \sigma \ln\left(\frac{W_N}{W_F}\right), \quad \sigma > 0$$

$$\Rightarrow \Delta \ln\left(\frac{W_N}{W_F}\right) = -\frac{1}{\sigma} \Delta \ln\left(\frac{L_N}{L_F}\right)$$

e.g. $L_F \uparrow 12\%$

	W_N/W_F
$\sigma = 4$	$\uparrow 3\%$
$\sigma = 12$	$\uparrow 1\%$
$\sigma \to \infty$	0%

→ relative wage $= \frac{MPL_N}{MPL_F}$ (Both MPL could fall due to more immigrants)

→ Why? firms choose the lowest wage

($\min\{a_N W_N, a_F W_F\} \Rightarrow a_N W_N = a_F W_F \Rightarrow \frac{W_N}{W_F} = \frac{a_F}{a_N}$, a_N constant)

· Immigration and Economic Growth

$$Y = A K^{\frac{1}{3}} L_s^{\frac{1}{3}} L_u^{\frac{1}{3}}$$

Suppose $L_s = L_{N,s} + L_{F,s} = 80 + 20 = 100$

$L_u = L_{N,u} + L_{F,u} = 180 + 20 = 200$

Assume perfect substitution within a skilled group, while complementation across skill groups.

Growth rate function:

$$g_Y = g_A + \frac{1}{3} g_K + \frac{1}{3} g_s + \frac{1}{3} g_u$$

Now, two skilled immigrants and four unskilled immigrants come in

$$g_s = \frac{2}{100} = 2\% \quad , \quad g_u = \frac{4}{200} = 2\%$$

Direct contribution:

$$\frac{1}{3} g_s + \frac{1}{3} g_u = \frac{1}{3}(2\%) + \frac{1}{3}(2\%) = \boxed{1.33\%}$$

Indirect contribution:

$$r = MPK = \frac{1}{3} K^{-\frac{2}{3}} L_s^{\frac{1}{3}} L_u^{\frac{1}{3}} \Rightarrow g_r > g_Y \Rightarrow K \uparrow \text{ until } g_K = g_Y$$

When $g_K = g_Y$, $g_r = 0$

$$0 = -\frac{2}{3} g_K + \frac{1}{3} g_s + \frac{1}{3} g_u \Rightarrow g_K = g_Y = \frac{\frac{1}{3}(2\%) + \frac{1}{3}(2\%)}{\frac{2}{3}} = 2\%$$

$\Rightarrow$ Indirect contribution: $2\% - 1.33\% = \boxed{0.67\%}$